Enrollment No: _____

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech Mid Semester Exam-WINTER 2025-26

Semester:6th | **Date: (19/02/2026)**
Subject Code: 303105349 | **Time: 09:45 AM -11:15 AM**
Subject Name: Compiler Design | **Total Marks: 40**

Instructions:

- Figures to the right indicate full marks.
- Que:1 is compulsory.
- Draw neat and clean drawings& Assume suitable data if necessary.
- BT- Blooms Taxonomy Levels -- Remember-1,Understand -2,Apply-3, Analyse-4,Evaluate-5, Create-6

			M	CO	BT
Q.1	(A)	**Five One line Questions (Each 1 Mark)**	05		
		1. What is the output of syntax analysis phase?		1	1
		2. Which sets are required to construct predictive parsing table?		3	2
		3. Name the parser that uses one symbol lookahead.		4	1
		4. Which tool is used to generate parsers automatically?		1	1
		5. What type of parsing is shift-reduce parsing?		2	1
	(B)	**Five Fill in the Blanks (Each 1 Mark)**	05		
		1. In LR parsing, the substring that matches RHS of production is called _____.		1	1
		2. Ambiguity in grammar means more than one _____ tree for a string.		1	2
		3. The technique used to remove ambiguity in grammar is _____.		3	2
		4. The tool used to automatically generate lexical analyzer is _____.		3	1
		5. LR parser performs _____ derivation in reverse.		1	2
Q.2		**Attempt any four Questions ()**	12		
	1.	Explain the term pattern, lexeme, token.		3	2
	2.	Explain the concept of Left Recursion and why it must be removed in top-down parsing.		2	3
	3.	Compute FIRST for the following grammar:		2	3

S ->ACB | CbB | Ba
A -> da | BC
B -> g | ϵ

C -> h | ϵ

	4.	Convert the following grammar to left factored form:		4	3
		S -> aSSbS | aSaSb |abb | b			
	5.	What is Lex? Explain the function of lex with structure.		2	4
Q.3		**Attempt any two Questions (Each 04 Marks)**	08		
	1.	Explain ambiguous grammar and parse tree. With Example.		2	3
	2.	What is Handle and Handle Pruning? Illustrate the process using an example.		2	3
		S → aAB			
		A → b			
		B → c			
		Input: abc			
	3.	Define LR(0). Illustrate the given grammar .Check the grammar is LR(0) or not.		3	4
		S -> AA			
		A -> aA | b			
Q.4	(A)	Define SLR(1). Illustrate the given grammar .Check the grammar is SLR(1).	05	5	4
		S -> cAd			
		A -> ab | e			
	(B)	Explain the structure of a compiler with a block diagram.	05	3	3
		OR			
	(B)	Explain Recursive descent Parser and LL(1) with components.	05	4	4